THE NEW DNA DIETS

Roll over nutrition fads. Nutrigenomics now provides hyperpersonalised diets based on one's genetic makeup and offers clues to prevent and treat disease well in advance

By SONALI ACHARJEE / *Graphics by* ***TANMOY CHAKRABORTY***

IN **THEIR NEVER-ENDING QUEST FOR HEALTHIER FOODS,** along with the holy grail of losing weight, humans in the modern era have taken refuge in dieting trends. Fads like the Mediterranean diet, or the Vegan diet, are everywhere, each involving subtracting or adding a nutrient—no fat, more fat, no sugar, fasting, no carb, more carb, no meat.... This approach in recent years has evolved into what is now called 'biohacking'—the mostly DIY practice of using scientific evidence to modify one's lifestyle for better health.

Now, biohacking has taken a far more serious turn from fancy diets, and it emerges from the depths of biological/ medical science. Incredible advancements in genetic diagnostics have opened up a whole new field called nutrigenomics. This discipline focuses on how food interacts with people's genes, how genes affect the body's response to food and thus their health. It also seeks to find new avenues to prevent and treat disease. Nutrigenomics allows for hyperpersonalised biohacking, tailoring nutrition plans based on an individual's genetic profile. A person may thus be advised to avoid dairy because her genes indicate lactose intolerance. Another can be encouraged to eat rice because there is medical proof to show her body uses the cereal better.

THE WORLD OF NUTRIGENOMICS

The interplay of genes and health outcomes has been theoretically known for over a decade. According to a 2013 study in the journal, *Nature Reviews Genetics*, nutrients can modulate gene expression—the process by which instructions encoded within a gene are used to create proteins crucial for cell function—through mechanisms like DNA methylation (where methyl groups, a fundamental building block molecule, is added to the DNA, thus altering gene expression). The study of these mechanisms and changes in gene expression through nutrients and medicines—without alterations to the underlying DNA sequence—is called epigenetics. Changes in gene expression influence how genes are read and used by cells, thus affecting biological processes. These changes not only have a bearing on disease susceptibility but also influence the effectiveness of dietary interventions.

Over the past decade, advances in DNA sequencing technologies like next-generation sequencing (NGS) and exome sequencing—where protein-coding regions (exons) of a genome is analysed—have improved the accuracy of nutrigenomic testing. These tests assess gene variants that influence the body's response to nutrients, metabolism and susceptibility to diseases. They point out allergies and nutrient deficiencies, and also look at how a body absorbs, metabolises and uses a nutrient. It identifies predispositions to conditions like gluten sensitivity or slower caffeine metabolism, not through allergy markers but through genetic traits. According to market re-



"Over 10,000 people have tried EPLIMO (epigenetic lifestyle modification). We have seen amazing improvements in their health markers after three months. This proves that biohacking is measurable"

SAJEEV NAIR, Founder and chairman, Vieroots